Question 1. Multivariate AR(1): Covariance and Stability

Consider

$$\mathbf{x}_{n+1} = \mathbf{A}\mathbf{x}_n + \boldsymbol{\eta}_n, \qquad \mathbf{x}_n \in \mathbb{R}^2,$$

with independent zero-mean Gaussian innovations, independent of the current state and its past. Their proposed covariance is

$$\mathbf{Q} = \mathbf{I}_2 - \mathbf{A}\mathbf{A}^\mathsf{T}.$$

Assume a zero-mean initial state with finite covariance, independent of future innovations. A covariance matrix must be positive semidefinite.

(a) Noise covariance and stability. **[20 points]** Using the singular value decomposition

$$\mathbf{A} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\mathsf{T}, \qquad \boldsymbol{\Sigma} = \mathrm{diag}(s_1, s_2),$$

derive $\mathbf{Q}$ in terms of $\mathbf{U}$ and $\boldsymbol{\Sigma}$. State the condition on the singular values for $\mathbf{Q}$ to be admissible, and the condition on the eigenvalues of $\mathbf{A}$ for asymptotic stability. Does admissibility alone guarantee asymptotic stability?

(b) Covariance balance and a marginal mode. **[20 points]** Derive

$$\mathbf{P}_{n+1} = \mathbf{A}\mathbf{P}_n\mathbf{A}^\mathsf{T} + \mathbf{Q}, \qquad \mathbf{P}_n = \mathbb{E}[\mathbf{x}_n\mathbf{x}_n^\mathsf{T}],$$

and verify that $\mathbf{P} = \mathbf{I}_2$ satisfies the stationary covariance equation whenever $\mathbf{Q}$ is admissible.

For $\mathbf{A} = \mathrm{diag}(1, 0.5)$, determine whether initial perturbations decay and whether $\sup_n \mathbb{E}\|\mathbf{x}_n\|_2^2 < \infty$. If $\mathbf{x}_0 \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_2)$, is the process stationary? Explain the distinction between stationarity and asymptotic stability.

(c) Unbounded variance. **[10 points]** Let $\mathbf{A} = \mathrm{diag}(1.1, 0.5)$. Explain why $\mathbf{I}_2 - \mathbf{A}\mathbf{A}^\mathsf{T}$ is not a valid covariance.

Instead prescribe $\mathbf{Q} = \mathbf{I}_2$ and $\mathbf{x}_0 = \mathbf{0}$. Derive $\mathrm{Var}(X_{1,n})$ and show that it diverges. Briefly distinguish unbounded variance over time from the unbounded support of a Gaussian distribution.

Question 2. One-dimensional DDPM

Let x_0 be a scalar sample from the bimodal data distribution

$$q_{\text{data}}(x) = \tfrac{1}{2}\mathcal{N}(x; -2, 0.25) + \tfrac{1}{2}\mathcal{N}(x; 2, 0.25),$$

where $\mathcal{N}(x; \mu, v)$ denotes a Gaussian density with mean μ and variance v. A denoising diffusion probabilistic model (DDPM) uses the forward process

$$x_t = \sqrt{\alpha_t}\, x_{t-1} + \sqrt{\beta_t}\, \xi_t, \qquad \alpha_t = 1 - \beta_t, \qquad \bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s,$$

where $0 < \beta_t < 1$ and $\xi_t \overset{\text{iid}}{\sim} \mathcal{N}(0, 1)$ are independent of x_0. The index $t = 1, \ldots, T$ denotes a diffusion step.

(a) Forward distribution. [15 points] Derive the conditional distribution $q(x_t \mid x_0)$ and show that x_t can be sampled directly using

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1 - \bar{\alpha}_t}\, \epsilon, \qquad \epsilon \sim \mathcal{N}(0, 1).$$

For fixed x_0, what happens to this distribution as $\bar{\alpha}_t \to 0$?

(b) Training target and conditional probabilities. [20 points] A neural network $\epsilon_\theta(x_t, t)$, with trainable parameters θ, predicts noise from x_t and t. Write its mean squared error training objective, with $x_0 \sim q_{\text{data}}$, $t \sim \text{Uniform}\{1, \ldots, T\}$, and $\epsilon \sim \mathcal{N}(0, 1)$ sampled independently. Does its target ϵ represent the noise added only during $t - 1 \to t$, or the standardized accumulated noise in part (a)?

For $x_0 = 2$, $\bar{\alpha}_t = 0.64$, and $\epsilon = -1$, calculate x_t. If $\epsilon_\theta(x_t, t) = -0.8$, calculate the squared error. Explain the distinction between $q(x_{t-1} \mid x_t, x_0)$ and $p_\theta(x_{t-1} \mid x_t)$. Why can generation not require the original x_0 as input?

(c) Reverse sampling. [15 points] Use the supplied reverse mean

$$\mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}} \left[x_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(x_t, t) \right].$$

For $t > 1$, sample

$$x_{t-1} = \mu_\theta(x_t, t) + \sigma_t z_t, \qquad z_t \sim \mathcal{N}(0, 1),$$

where σ_t is a prescribed reverse standard deviation and the z_t are fresh independent draws. At $t = 1$, return $\mu_\theta(x_1, 1)$ without adding noise. Assuming $\bar{\alpha}_T$ is sufficiently small, write pseudocode for generating one new sample, including its starting distribution. Explain why generation does not retrace the exact forward noise realization of a training sample.